

ASSIGNMENTS – OOADI

LECTURE 9

1. Split your train program with the two types of threads into two separate programs (projects). One is the client program which contains the user input and one is the server program which handles the list of trains.
2. Extend the train list server to also from time to time write the list to a file. If we wish to restart the server part, we should also read the file and initiate the list with the saved list.
3. Extend the client/server to allow the interaction to happen across a network. Use your local network and you fellow students' laptop to run. You may need to open a port on the firewalls of the machines involved. Feel free to use a virtual machine or two instead as an alternative, or just use the internal loopback interface (127.0.0.1) and start the client/server as individual programs.
 - a. First use the client with manual inputs/edits.
 - b. Then extend the client design and implementation to trigger a random action based upon a mouse click.
 - i. How fast can you "click" a request before you run into some issues? What type of issues to you experience? Can you do something about this?
 - c. Then use the client with automatic actions i.e. the client should randomly without user interaction select a query, send it to the server and show the response.
 - i. How fast can you have a large (as many as feasible) number of client random queries executed, as fast as possible, without running into problems? What kind of problems do you observe? What can you do about these problems?
 - ii. Does it matter if you use a TCP or UDP socket for the communication?